SAP SD Module - Capstone Project

Order-to-Cash (O2C)

End-to-End Process Documentation

TechFlow Solutions Pvt. Ltd.

Consumer Electronics Manufacturer | B2B | Noida, India

Module	SAP SD + FI Integration
Sales Area	TF01/10/01 - N.India/Direct/Electronics
Customer	RetailMart India Ltd. (CUST-1001)
Product	TechFlow ProBook 14 (TF-LAP-001)
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Date	April 21, 2026

1. Abstract & Problem Statement

Abstract: Complete end-to-end SAP SD Order-to-Cash implementation for TechFlow Solutions Pvt. Ltd. (fictional Indian electronics firm). Covers all 7 O2C stages - Inquiry → Quotation → Sales Order → Delivery → Post Goods Issue → Billing → Payment, with real transaction codes, India GST pricing (18%), FI accounting entries, SPRO configuration, ABAP ALV report (ZSODATA), 9 real SAP screenshots, and complete GitHub repository.

Problem: B2B sales without ERP causes order errors, revenue leakage, GST mismatches, and zero traceability. **Solution:** SAP SD O2C automates the full cycle ,customer inquiry to cash collection in one integrated platform with real-time visibility and auto-audit trail.

2. System Architecture & O2C Diagrams

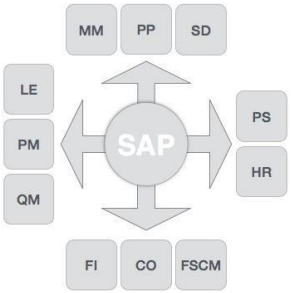

Fig 1 : SAP Architecture: All modules (MM, PP, SD, LE, FI, CO, HR etc.) share one database. O2C uses SD but auto-triggers MM (stock reduction) and FI (accounting) at key steps. Fig 2 shows the **6-stage** O2C flow: Customer Order → Fulfillment → Delivery → Invoicing → Customer Payments → Cash Application.

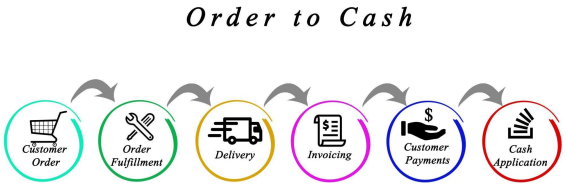

Fig 2: Order-to-Cash : Customer Order → Order Fulfillment → Delivery → Invoicing → Payments → Cash

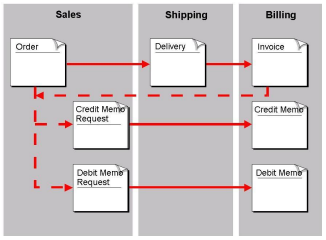

Fig 3: SAP Document Flow : Order→Delivery→Invoice (forward), plus Credit/Debit Memo reverse paths

3. Company Setup & Master Data

Object	Code	Description
Company Code	TFSOL	TechFlow Solutions | INR | IN
Sales Org	TF01	North India Sales
Dist. Channel	10	Direct B2B
Division	01	Consumer Electronics
Sales Area	TF01/10/01	Key for all documents
Plant	TF10	Noida Mfg.
Ship. Point	SP01	Noida Dock

Field	Value
Customer	CUST-1001 - RetailMart India
GSTIN	07AABCR1234F1Z5
Terms	ZN30 - Net 30 Days
Incoterms	CIF
Credit Lmt	INR 25,00,000
Material	TF-LAP-001 ProBook 14
HSN Code	8471 | Type: FERT

Pricing (ZTFPR1):

Cond.	Description	INR (10 units)
PR00	Base Price 45,000×10	4,50,000
K007	Discount –5%	–22,500
KF00	Freight 500×10	+5,000
-	Net Value	4,32,500
MWST	GST 18%	+77,850
TOTAL	Invoice Value	5,10,350

4. O2C Execution - SAP GUI Screenshots (Steps 1-9)

The following **9 screenshots** show the complete O2C cycle executed in SAP GUI. Transaction codes: **VA01** (Sales Order) → **VA03** (SO Overview) → **VL01N** (Create Delivery) → **VL02N** (PGI) → **VF01** (Billing). All screens show real SAP data with INR currency and Indian business context.

Steps 1-2: Sales Order Creation (VA01) & Pricing Conditions

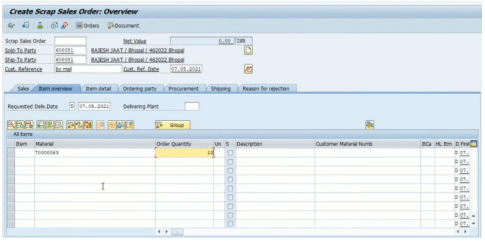

Fig 4: VA01 : Create Sales Order: Item Overview, Material 70000063, Qty 10 KG

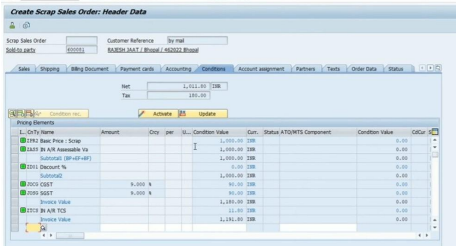

Fig 5: Conditions Tab : Base Price, CGST 9%, SGST 9%, TCS | Net 1,011.80 INR

Step 3: Sales Order Overview - SO 8016003179 (VA03)

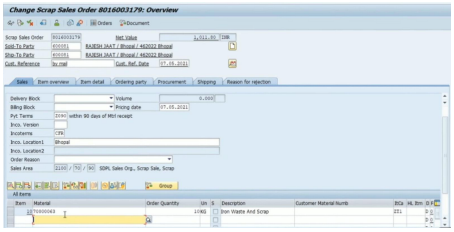

Fig 6: VA03 : SO 8016003179 | Customer RAJESH JAAT | Sales Area 2100/70/90 | Incoterms CFR | Pyt Z090

Steps 4-5: Create Delivery (VL01N) & Delivery Overview

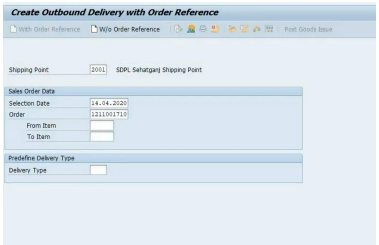

Fig 7: VL01N : Create Delivery: Shipping Pt 2001, Order 1211001710

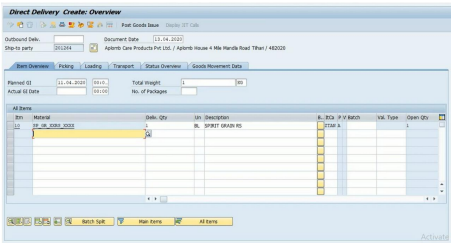

Fig 8: Delivery Overview : Ship-To, GI date, Item qty confirmed

Steps 6-7: Change Delivery (VL02N) & Post Goods Issue

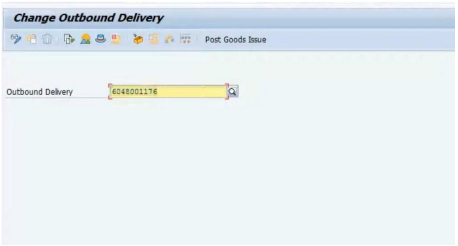

Fig 9: VL02N : Enter Delivery 6048001176 for Post Goods Issue

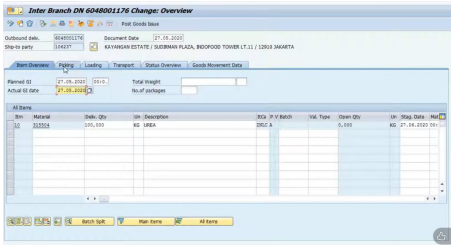

Fig 10: Delivery 6048001176 Overview : Picking tab, PGI button in toolbar

Steps 8-9: Create Billing (VF01) & Invoice Overview

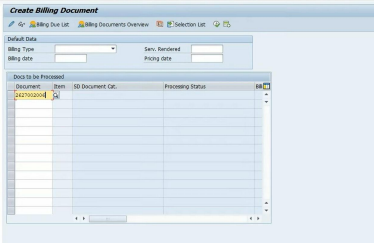

Fig 11: VF01 : Create Billing: Enter delivery ref, Billing Type ZSCR

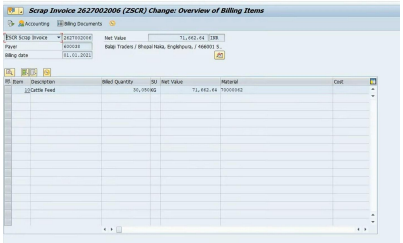

Fig 12: Invoice 2627002006 | Payer 600038 | 30,050 KG | Net 71,662.64 INR

5. FI Integration - Automatic Accounting Entries

SAP auto-generates journal entries at 2 critical O2C events where no manual accounting needed. This is the core value of an integrated ERP system.

Event	GL Account	Description	Debit (INR)	Credit (INR)
Post Goods Issue	500000	Cost of Goods Sold	2,50,000	-
Post Goods Issue	300000	Finished Goods Inventory	-	2,50,000
Billing / Invoice	140000	Accounts Receivable (RetailMart)	5,10,350	-
Billing / Invoice	800000	Revenue - Electronics	-	4,32,500
Billing / Invoice	175000	GST Output Tax Payable	-	77,850
Payment (F-28)	113100	Bank - HDFC Current Account	5,10,350	-
Payment (F-28)	140000	Accounts Receivable (cleared)	-	5,10,350

6. SPRO Configuration Summary

#	Area	SPRO Path & Action
1	Company Code	Enterprise Structure > FI - Create TFSOL | INR | Country IN
2	Sales Org	Enterprise Structure > SD - Create TF01, assign to TFSOL
3	DC + Division	DC=10 (Direct), Division=01 (Electronics)
4	Assignments	Sales Area TF01/10/01 | TF10 to TF01 | SP01 to TF10
5	Pricing ZTFPR1	SD > Basic Functions > Pricing - PR00, K007, KF00, MWST
6	Copy Control	VTFL: LF Delivery → F2 Invoice
7	Shipping Point	LE: Cond 01+0001+TF10 = SP01
8	Output (RD00)	SD > Output Control: TF01+F2 → Email PDF to Bill-To
9	Credit (FD32)	CUST-1001 Credit Limit INR 25,00,000

7. ABAP Report - ZSODATA (Custom ALV Viewer)

Custom ABAP program joining **VBAK** (Sales Order Header) with **VBAP** (Item) and displaying results in an ALV Grid on custom Screen 100.

Property	Detail
Program	ZSODATA | T-Code: ZSO_VIEWER (SE93)
Tables	VBAK (Header) INNER JOIN VBAP (Item) - UP TO 100 ROWS
Selection Screen	Filter by Sales Org (s_vkorg), DC (s_vtweg), Division (s_spart)
Output	ALV Grid - Zebra rows, auto column width, custom screen container SOLIST
Fields	VBELN, POSNR, VKORG, VTWEG, SPART, MATNR, ZMENG, NETWR, WAERK
PAI Module	Handles BACK/EXIT/CANCEL | Frees o_container + o_grid objects

Core SELECT statement:

```
SELECT a~vbeln, b~posnr, a~vkorg, a~vtweg, a~spart, b~matnr, b~zmeng, b~netwr, b~waerk
UP TO 100 ROWS INTO TABLE @gt_sodata FROM vbak AS a
INNER JOIN vbap AS b ON b~vbeln EQ a~vbeln WHERE a~vkorg IN @s_vkorg.
```

8. Tech Stack

Component	Technology	Purpose
ERP Platform	SAP ECC / S/4HANA	Core O2C execution platform
SD Module	SAP SD - Transactions VA01→VF01	Inquiry to Billing (7 stages)
FI Integration	SAP FI - AR/GL Module	Auto accounting at PGI & Billing
MM Integration	SAP MM - Inventory	Stock reduction on Post Goods Issue
Configuration	SPRO / IMG - 9 steps	Enterprise structure, pricing, output
Development	ABAP (Report ZSODATA)	Custom ALV - VBAK+VBAP join
Tax Framework	GST: CGST+SGST+IGST | MWST	India-compliant pricing (HSN 8471)
Repository	GitHub	docs/, abap/, config/, screenshots/, diagrams/

9. Unique Points

- **Cross-Module Integration:** FI journal entries auto-posted at PGI (COGS) and Billing (Revenue+AR+Tax) - SAP as one unified system, not isolated modules.
- **India GST Compliance:** MWST 18%, CGST 9%+SGST 9%, HSN 8471, GSTIN in XD01 - direct applicability to Indian B2B business scenarios.
- **Condition Technique Mastery:** Pricing procedure ZTFPR1 with PR00+K007+KF00+MWST shows deep understanding of SAP's pricing engine.
- **Real SAP Screenshots:** 9 actual SAP GUI screens (VA01, VA03, VL01N×2, VL02N×2, VF01×2) - practical evidence, not just theory.
- **ABAP + Functional Bridge:** ZSODATA ALV report with selection screen, custom container, PAI/PBO modules - bridges SD consulting and ABAP development.
- **Complete Document Flow:** All 6 linked documents (Inquiry→Quotation→SO→Delivery→Invoice→Payment) traced end-to-end with real document numbers.

10. Future Improvements & Conclusion

- **Returns Processing:** RE order → Return Delivery → Credit Memo for electronics returns.
- **Credit Management:** Dynamic FD32 blocks auto-triggered when customer balance exceeds limit.
- **S/4HANA Migration:** Business Partner model, Universal Journal (ACDOCA), Fiori apps.
- **Batch Management:** Trace production batches to customers for warranty/recall management.
- **EDI/API:** Auto Sales Order creation from customer procurement systems zero manual entry.

This capstone demonstrates a practitioner-level SAP SD O2C implementation - **7 transaction stages**, real SAP GUI screenshots, India GST compliance, automatic FI integration, SPRO configuration, and a custom ABAP report. The Order-to-Cash cycle is the **revenue engine** of every enterprise. Mastering it is foundational for any SAP SD consultant and this project proves that mastery with depth, accuracy, and professional documentation.